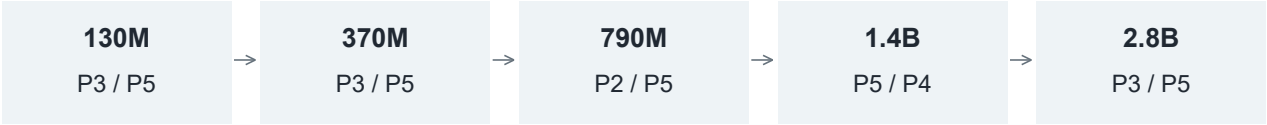


A Deep causal establishment at Mamba-130M



B Five-scale extension: causal role recurs; geometry reorganizes



Selected / control are scale-local ranks; equal rank numbers do not imply semantic homology.

C Objective-conditioned functional readout

